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Paleomagnetic Results from Some Permian-Triassic Rocks from Southwestern China

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Abstract. The Ailao Shan fault zone in West Yunnan represents a suture boundary between South China and Indosinian Cratons. Preliminary paleomagnetic results from three sites in Sichuan, Guichou, and Yunnan Provinces reveal a 14° northward migration of South China Craton between Late Permian and Middle Triassic. The two cratons were separated by about 20° latitudes during the Early Triassic. The tectonic setting of East Asia during the Triassic was probably analogous to the present Indonesian region.

Introduction

The Ailao Shan fault zone (West Yunnan, China) extends from the eastern margin of Tibet Plateau along the Red River into the Gulf of Tongking in the southeast. The fault zone is a structural and stratigraphic boundary between Indosinian Craton in the west and South China Craton in the east. The fossil assemblages found in Indosinian Craton are remarkably different from those found in South China Craton. Permian-Triassic fossils found in Indosinian Craton are similar to the Gondwana fossil assemblages, suggesting that the craton was probably part of Gondwanaland during that period of time (Tarling, 1971; Stauffer, 1974; Acharyya, 1978). In South China Craton, Gondwana-type fossil assemblages are only present in rocks older than the Permian-Carboniferous (Wang and Liu, 1979). Little relation between the Middle-Upper Triassic fossil assemblages of South China Craton and those of Indosinian Craton has been reported (Yin, 1980).

A series of ultramafic bodies, generally sheared and serpentinized, are exposed on the western side of the fault zone. Glaucophane, pumpellyite, stilpnomelane, and other minerals indicative of high pressure/temperature ratio are present in these rocks. The entire association is interpreted as an ophiolite melange (Bally et al., 1980). Emplacement of plutons occurred along the fault zone during two time intervals (Geol. Acad. China, 1973). The earlier intrusions, which are mainly granite, porphyry,and granodiorite bodies, occurred during the Middle-Late Triassic Indosinian Orogeny. Near the cities of Jinping and Yuanjiang (West Yunnan), the intrusive bodies are generally covered by Upper Triassic sediments. The later intrusions occurred during the Jurassic Yenshanian Orogeny. Isotopic dating of rocks belonging to the later intrusions indicates an age of 111 m.y. (Geol. Acad. China, 1973). The geological setting of the area suggests the presence of a subduction zone during the early Mesozoic.

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The time of collision of Indosinian Craton with South China Craton has not been well determined. The Upper Triassic formations on the northern side of the Ailao Shan fault consist predominantly of coal-bearing shales and clastic, continental-facies sedimentary rocks. The Triassic rocks on the southwestern side of the fault zone are mainly marine or littoral deposits. A series of Norian-age red beds, however, is present near the city of Mojiang. Misch (1947) suggested that the Triassic formations on the southwest side of the fault were derived from Indosinian Craton while those on the northwest side were derived from South China Craton. The contrasting geological and stratigraphic characteristics of the two cratons have led us to believe that they were separated by a great distance during the late Paleozoic through early Mesozoic. This hypothesis has not been corroborated by studies of plate motions due to the scanty paleomagnetic coverage of this area. In an attempt to obtain fundamental data for reconstructing the tectonic history of the area, we have carried out a paleomagnetic study of several Permian and Triassic formations near the Ailao Shan fault zone.

Sampling Sites

We visited three geological sections in 1980. The first section is an exposure of the Iwanshui Formation near the city of Mojiang located 10 km west of the fault zone in West Yunnan. The Iwanshui Formation is a 1500 m series of reddish silty shales overlying the ultramafic bodies. The section is dipping at 25-30° to the east. The presence of Barmesia lirata and Halobia yunnanensis in the sediments indicates an Early Triassic age. We have collected oriented cores at four sites of the section.

The second site is an exposure of the Huachi Formation near the city of Guiyang in Guichou Province. The formation is a series of thickly-bedded dolomitic limestones with few fossils. The overlying sediments, however, contain Carnian Paleoneilo lineata, Gervilleia cf. angusta, G. cf. musculosa, and Myophoria cf. acuticostata while the underlying clastic sediments contain Anisian Nuculana cf. perlonga, N. timorensis, Cassianella sp., and Enteropleona sp. Hence, the Huachi Formation was deposited during Carnian to Anisian time. Oriented cores were collected from both limbs of an open anticline; therefore, we could apply a fold test to the paleomagnetic orientations of the samples.

The third sample site is a Permian-Triassic section near the city of Omei in Sichuan Province. The Upper Permian of the section is a thick series of basalts and andesitic basalt flows, occasionally intercalated by limestones. The total thickness of this series of basalts, known as the Omeishan Basalts, exceeds 3000 m at some places. The Omeishan Basalts are overlain by 200 m limestones and coal-bearing siltstones. McElhinny

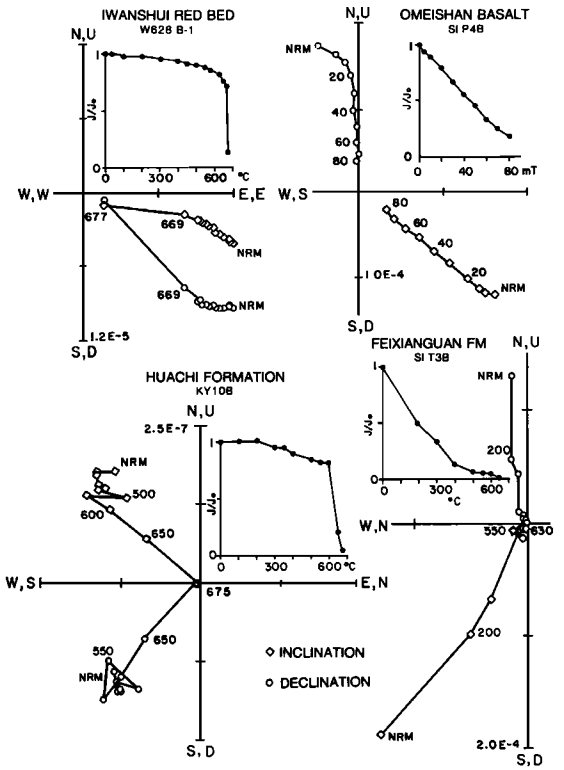


Fig. 1. Orthogonal projections of demagnetization vectors of typical specimens from the formations studied.

et al. (1981) have reported some paleomagnetic results of the Omeishan Basalts from the same site. Their results indicate a paleolatitude close to the equator of the site during the Permian. Overlying the Omeishan Basalts is the Feixianguan Formation, which consists of 330 m of reddish sandstones, siltstones, and limestones. Fossils found in this formation include Claraia claraia, Eumorphotis inaequicostata (Beneck), and Entolium descites (Schlotheism), which are indicative of an Early Triassic age. The section we visited is part of an overturned anticline dipping between 50-80°. Oriented block samples were collected from both the Omeishan Basalts and the Feixianguan Formation.

Paleomagnetic Results

The samples from Guiyang and Sichuan were treated with thermal demagnetization up to 675°C at increments of 50°C and with alternating field demagnetization up to 80 milliteslas (mT) at increments of 10 mT. Generally, thermal demagnetization is more efficient than af demagnetization for cleaning the secondary magnetic overprints in the specimens.

The orthogonal vector projection of the demagnetization curves reveals the presence of at least two magnetic components in the specimen of the Iwanshui Formation (Fig. 1). The magnetic component isolated at about 200°C has blocking temperatures close to 675°C. The marked decrease of magnetic intensity at 669°C indicates that hematite is the magnetic carrier of this component. We have chosen the magnetic direction at 669°C as the stable direction for the specimens. The mean direction of the four specimens is D=133.6°, I=16.3°, with an α_{95}=7.1° (Fig. 2). Assuming a reversed polarity during this period, the virtual geomagnetic north pole (VGP) is located at 34.6°N, 17.6°W.

Ten samples of the Huachi Formation have been measured. We have deleted two samples from our consideration because they show very soft and unstable magnetic directions at temperatures higher than 100°C. A characteristic direction is isolated at 100-200°C in the other eight specimens (Fig. 1). The specimens show a maximum blocking temperature of about 670°C, indicating the presence of hematite as the magnetic carrier. We have chosen the direction at 200°C to represent the stable direction for each specimen. The in-situ directions of the eight accepted specimens show a great dispersion (Fig. 2). After applying the bedding correction, the magnetic directions cluster generally toward the southwest, indicating a successful fold test. One specimen shows a normal inclination and a northeastern declination nearly antiparallel to the other seven specimens. The results suggest that the Huachi Formation carries a stable magnetism acquired shortly after deposition. The mean direction of the eight samples is D=215.7°, I= -26.1°, with an α_{95}=5.7°. The VGP is located at 54.6°N and 150.9°W.

Thirteen basalt and andesite flows and one limestone layer in the Omei section were sampled. We have found that the volcanic rocks containing large vesicules filled with dark-colored secondary minerals generally show unstable magnetic directions. A stable direction, however, is isolated at 10-20 mT in six specimens and the limestone (Fig. 1). Alternating field demagnetization is generally sufficient for cleaning the magnetic overprints in these specimens. The mean direction of the seven accepted samples from this section is D=22.9°, I= 2.2°, with an α_{95}=19.5° (Fig. 2).

We have also measured eight samples of the Feixianguan red beds (Fig. 1). The mean direction of the samples after thermal demagnetization at 500°C is D=42.8°, I= -18.3°, with an α_{95}=18.5°.

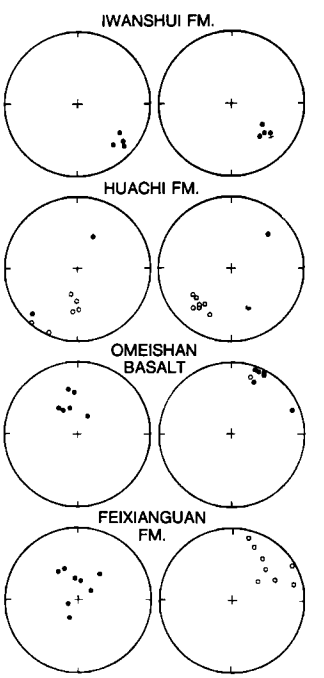


Fig. 2. Lower hemisphere projections of the stable directions determined for the specimens of the four formations.

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This particular demagnetization step is chosen according to the criteria of minimum dispersion. Since the sample sites were all located on one limb of an anticline, the scatter in the magnetic direction does not improve after applying the tilt correction. Also, the reddish color of the sediments and the high blocking temperatures of the magnetic components indicate the presence of hematite in the sediments; therefore, remagnetization of the sediments may have occurred. The mean in-situ directions of the specimens, however, differ significantly from the present geomagnetic field (Fig. 2). The magnetic directions obtained after the tilt-correction, on the other hand, show a declination similar to that of the underlying Permian basalts. Hence we believe that the acquisition of the NRM in the Feixianguan Formation predates the folding of the sediments.

Discussion

The paleomagnetic results presented above are preliminary since the sample sizes involved are generally small. Published paleomagnetic results from this area, however, are scanty at present. The main objective of presenting the preliminary results from this study here is to stimulate further research in this subject.

The successful fold test and the nearly antiparallel directions observed in the Huachi Formation suggest an early acquisition of the NRM. M. Fuller and J. Lin (pers. comm.), and N.O. Opdyke (pers. comm.) have independently obtained similar paleomagnetic directions for some Triassic rocks from south and southwestern China. The authors believe that the direction of the Huachi Formation represents the stable Middle-Triassic paleomagnetic direction of South China Craton. The sample site of the Huachi Formation is located within a series of open anticlines and synclines with no major thrust faults and is probably autochthonous relative to the basement rocks. Assuming a reversed geomagnetic polarity for the Huachi Formation, a paleolatitude of 14°N during the Middle Triassic is implied.

The paleomagnetic direction of the Permian Omeishan Basalts obtained from the present study is similar to that obtained by McElhinny et al. (1981). The data show a declination of 22° and an inclination of 2.2°, suggesting that the sample site was located at a paleolatitude close to the equator during the Late Permian. At present we are not certain about the geomagnetic polarity for the Omeishan Basalts. If they were formed during the Permian Kiaman Reversed period (Irving and Parry, 1963), a 180° finite rotation of the sample site must be invoked to explain the paleomagnetic declination.

The Lower Triassic Feixianguan Formation overlying the Permian basalts shows a declination similar to that of the basalts, but an inclination of -18°. If a normal polarity is assigned to the Feixianguan Formation, the sample site will be located at 9°S. This consideration implies a complicated pattern of plate motion of the South China plate--first southward from about the equator to the 9°S during the Late Permian-Early Triassic and then northward to 14°N during the Early-Middle Triassic. The estimated plate-motion rate is greater than 20 cm/m.y., which is unusually high.

We interpret that the Omeishan Basalts, the Feixianguan Formation, and the Huachi Formation

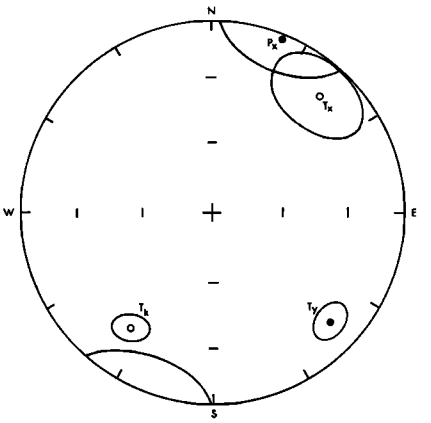


Fig. 3. Lower hemisphere projections of mean directions of the Omeishan Basalts (Px), Feixianguan Fm. (Tx), Huachi Fm. (Tk), and Iwanshui Fm. (Ty).

were all formed during reversed polarity intervals. A 9° northward migration of South China Craton occurred during the Early Triassic, resulting in an inclination of -18° for the Lower Triassic Feixianguan Formation, followed by a 5° northward migration during the Early-Middle Triassic, resulting in an inclination of -26° for the Middle Triassic Huachi Formation. The estimated plate-motion rate based on this consideration is about 9 cm/m.y., which is compatible with average present-day rates of plate motion. Accordingly, the sample sites of the Omeishan basalts and the Feixianguan Formation have undergone 180° rotation relative to the sample sites of the Huachi Formation at Guiyang. Such large rotations are possible since the former sites are located within the Lungmenshan thrust zone.

The paleomagnetic direction of the Iwanshui Formation west of the Ailao Shan fault zone differs significantly from that of the Huachi Formation in Guiyang (Fig. 3). Assuming a normal polarity for the Iwanshui red beds, the sample site was located at 13°S. We infer that Indosinian Craton was probably separated from South China Craton during the Early Triassic (Wang and Chan, 1981). The direction from the Iwanshui, however, is also different from previous results obtained for Southeast Asia (Fig. 4). If the sample site of the Iwanshui red beds is attached to Indosinian Craton, the craton itself was probably separated from the western part of Southeast Asia during the Triassic. Bunopas and Vella (1978) have suggested that the belt of ultramafic bodies in Central Thailand probably represents a major suture (The Nan Suture) between Shan-Thai Craton in the west and Indosinian Craton in the east. The two land masses probably underwent a different tectonic history before the Triassic.

Based on available paleomagnetic and geological evidence from Asia and China, we postulate a possible reconstruction of East Asia for the Early Triassic (Fig. 5). The paleolatitudes of the different land masses are based on the results from Haile (1981), McElhinny et al. (1981), and the present study. The paleolongitudes of the land masses are arbitrary; but the reconstruction is compatible with the idea that Indosinian Craton was close to Gondwanaland during the Early Triassic (Acharyya, 1981). Indosinian and South China

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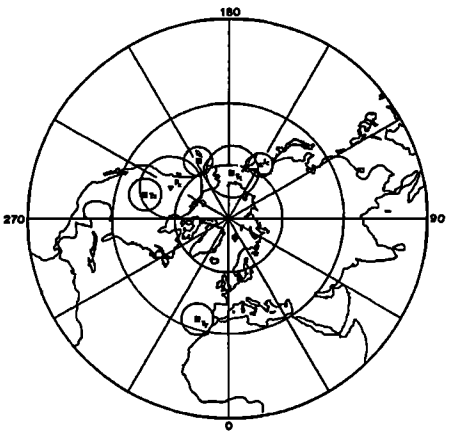


Fig. 4. Pole positions for different sites during the Mesozoic. Ty: Upper Triassic pole of Iwanshui Fm., West yunnan; Tk: Middle Triassic pole for Guiyang, South China; Tx: Lower Triassic pole of Feixianguan Fm., Sichuan; Px: Permian pole of Omeishan Basalts, Sichuan; Tt: Upper Triassic pole of west Thailand; and Jc: Jurassic pole of South China.

Cratons were separated by about 20° latitudes. Several land masses were present in the Paleotethys Ocean. The northern margin of the Paleotethys was probably a passive margin with thick 'miogeoclinal' deposits. The miogeoclinal sequence has now become the extensive, triangular-shaped belt of Triassic flysch, known as the Bayankara Group, between Northern Tibet and Sichuan. Based on stratigraphic evidence, Leith (1982) has suggested that a passive margin probably also existed west of Pamir.

An analogy of the tectonic setting of the area may be the modern-day Indonesian region. The passive margin north of the Paleotethys Ocean resembles the present South China Sea and the land masses in the Paleotethys Ocean resemble the Indonesian islands (Fig. 5). Collisions and fragmentations of these small blocks and land masses probably occurred before the major collision of India with Eurasia which has resulted in crumpling and smashing up of these blocks between the two major plates. During and before the collision, these separated blocks might have undergone relative rotations, resulting in large scatter of paleomagnetic directions over the area. The Nan Ophiolite in Central Thailand, the Song Ma Ophiolite in Northern Vietnam, and the ultramafic belts in Central Tibet and West Yunnan probably represent traces of narrow oceanic crust originally present between the land masses. The same scenario would probably occur in the future when Australia collides with South China and crumples up the Indonesian islands between the two land masses.

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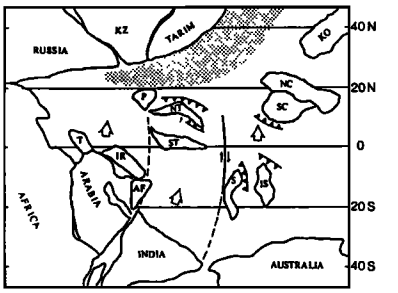


Fig. 5. A possible reconstruction of East Asia during the Middle Triassic. Open arrows indicate the general direction of plate motions. The shaded area represents the miogeoclinal sequence now represented by the Bayankara Group. The blocks shown are Kolyma (KO), North China Craton (NC), South China Craton (SC), Indosinian Craton (IS), Shan-Thai Craton (S), South Tibet (ST), North Tibet (NT), Pamir (P), Iran (IR), Turkey (T), Afganistan (AF), and Kazakhstania (KZ).

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